INCH-POUND

MIL-C-24643/29C
14 March 1994
SUPERSEDING
MIL-C-24643/29B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSMU
(INCLUDING VARIATION LSMA AND LSMUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in three variations: LSMU (unarmored), LSMA (armored) and LSMUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 20-7, maximum diameter 0.039 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
* Third - Ethylene propylene rubber or cross-linked polyethylene insulation white. Minimum average wall thickness 0.013 inch. Standard identification code applied by method 1.
* Fourth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, will-rounded assembly. Cabling sequence shall be consecutive starting with no. 1 from center outward.
Fifth - Binder tape, applied helically with overlap.

AMSC N/A FSC 6145

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Construction (non-watertight) (continued)

Construction variant type LSMU

* Sixth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

Construction variant for type LSMA

* Seventh- Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.
Eighth - Braided metal armor

Construction variant type LSMUS

* Sixth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Seventh- Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
* Eighth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Ninth - Binder tape applied helically with overlap.
* Tenth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

TABLE I. Details

Military 1/ part no. M24643/29	Size 2/	Number of Conductors	Diameter over Shielded core (inch nom.)	Overall Diameter Inches					
				TYPE LSMU		TYPE LSMA		TYPE LSMUS	
				min	max	min	max	min	max
-01UN	14	14	0.340	0.365	0.400	0.415	0.450	0.425	0.460

1/ Unarmored, overall shield. When armor option or double shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LSMU, LSMA and LSMUS.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum..	10.93
* Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Shield to shield (type LSMUS only, overall shields only).....	200
* Insulation resistance - megohms/1000 feet, minimum.....	
Conductor to conductor.....	2000
Shield to shield (type LSMUS only).....	100
* Jacket flaws (LSMUS only).....	No failure
* Conductor and shield continuity.....	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Group A:	
Visual and dimensional	No failure
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - 6 inch maximum mandrel diameter	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

Group C:	Requirements
Physicals - (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (conductor) - cycles minimum	25
* Permanence of printing (jacket) - (types LSMU and LSMUS only) cycles, minimum	125
Armor (type LSMA only) - conformance to material, construction and coverage	No failure
Shield - (type LSMUS only) conformance to material, construction and coverage	No failure
Surface transfer impedance (type LSMUS only) Milliohms per meter, maximum	700
Emp response time - db, minimum	60
Group D:	
* Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH:

2500 feet nominal.

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